

Graph Coloring

Time limit: 1.00 s

Memory limit: 512 MB

You are given a simple graph with n nodes and m edges. Your task is to use the minimum possible number of colors to color each node such that no edge connects two nodes of the same color.

Input

The first line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

Then there are m lines describing the edges. Each line has two integers a and b : there is an edge connecting nodes a and b .

Output

First, print an integer k : the minimum number of colors.

After this, print n integers $c_1, c_2, \dots, c_n$: the colors of the nodes. The colors should satisfy $1 \leq c_i \leq k$.

You may print any valid solution.

Constraints

- $1 \leq n \leq 16$
- $0 \leq m \leq \frac{n(n-1)}{2}$

Example

Input	Output
4 4 1 2 2 3 3 4 4 1	2 1 2 1 2